

Lab 6: Where Candidates Actually Go

84-355 Democracy's Data | Week 6

Your name here

August 11, 2026

Contents

Before we start	1
Part 1: Three hundred and seventy-seven days out	2
Part 2: Where they went	2
Part 3: Where they did not go	4
Part 4: Is it about size?	5
Part 5: Is it about closeness?	6
Part 6: Three things to notice before you trust any of this	8
Part 7: On your own	9
Where the data came from	14

Before we start

Chapter 5 is about campaign strategy — targeting, messaging, the allocation of a scarce resource. The scarcest resource a presidential campaign has is not money. It is **the candidate's time**. There are only so many days, and a candidate can be in exactly one place at a time.

So watch where they choose to stand. Every campaign stop is a decision, made under pressure, by people with the best polling in the country. Two hundred million Americans live somewhere the candidates never went.

Two weeks ago you found that only **8 states** were within five points in 2024 — 13 within ten. (`margin` in this file is **signed**: negative means Harris won. Test `abs(margin) < 5`, not `margin < 5`, or you will silently count every Democratic state as competitive and get 26.) Last week you found that PAC money goes almost entirely to people who already hold office. Today those two facts collect their consequence.

Everything is base R. Work in order.

Part 1: Three hundred and seventy-seven days out

```
v <- read.csv("data/campaign_visits_2024.csv", stringsAsFactors = FALSE)
st <- read.csv("data/pres2024_states.csv", stringsAsFactors = FALSE)
dim(v)
```

```
#> [1] 377 11
```

Every public campaign stop the Associated Press logged in 2024.

```
head(v[, c("date", "city", "state", "candidate", "event_type")], 5)
```

```
#>      date      city      state candidate      event_type
#> 1 2024-03-13 Milwaukee Wisconsin      Biden      Campaign
#> 2 2024-03-14  Saginaw  Michigan      Biden      Campaign
#> 3 2024-03-16  Vandalia    Ohio      Trump Support other candidate
#> 4 2024-03-19   Phoenix  Arizona      Biden      Campaign
#> 5 2024-03-19     Reno    Nevada      Biden      Campaign
```

```
table(v$candidate)
```

```
#>
#> Biden Harris Trump Vance Walz
#>   31    84   125    64    73
```

Five people: Biden until he withdrew in July, then Harris, plus the two running mates and Trump throughout.

```
v$date <- as.Date(v$date)
range(v$date)
```

```
#> [1] "2024-03-13" "2024-11-04"
```

March to the fourth of November — the day before the election.

Part 2: Where they went

```
visits <- as.data.frame(table(v$state))
names(visits) <- c("state", "visits")
visits$state <- as.character(visits$state)

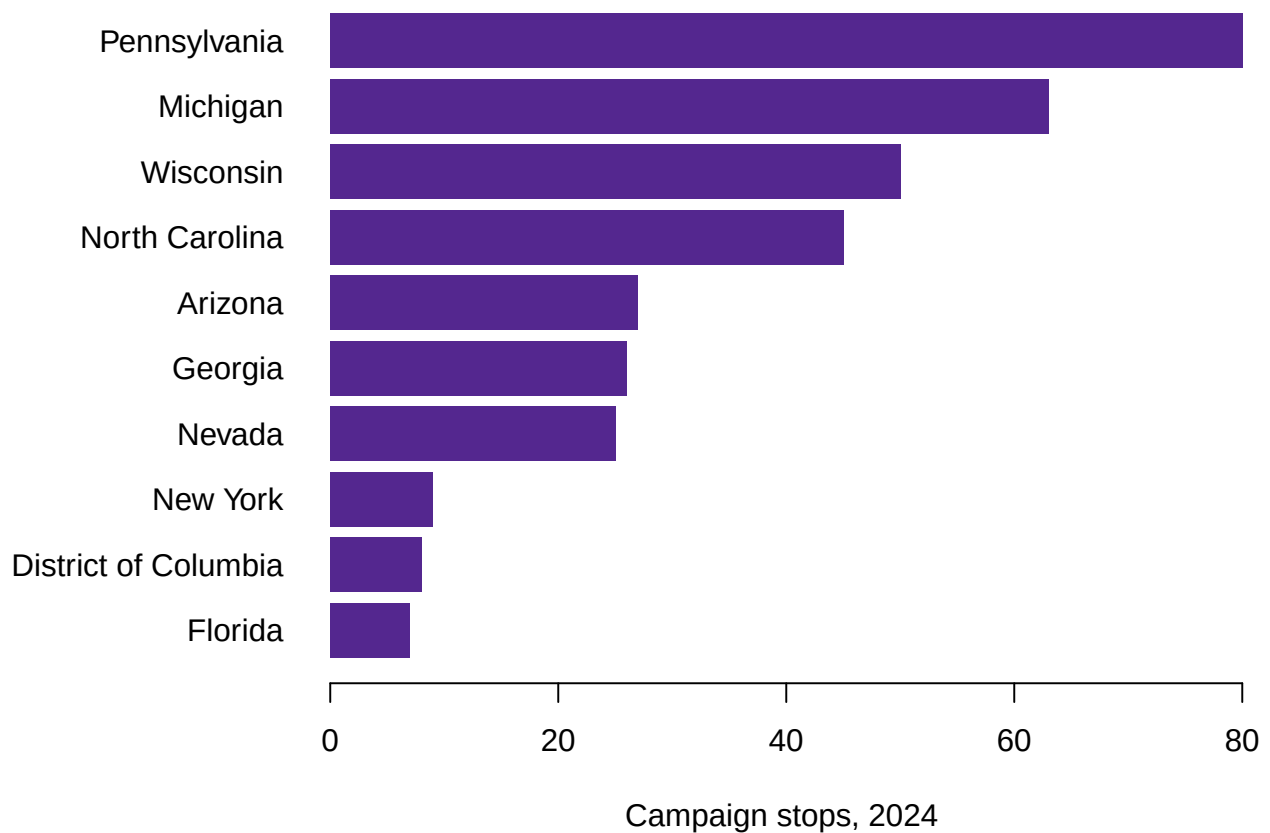
head(visits[order(-visits$visits), ], 8)
```

```
#>           state visits
#> 22  Pennsylvania     80
#> 11    Michigan      63
#> 26    Wisconsin     50
#> 20 North Carolina    45
#>  1     Arizona      27
#>  6     Georgia      26
#> 15     Nevada       25
#> 19     New York      9
```

```
top <- head(visits[order(-visits$visits), ], 10)
top <- top[order(top$visits), ]
```

```
par(mar = c(5, 11, 4, 2))
barplot(top$visits, names.arg = top$state, horiz = TRUE, las = 1,
        col = "#54278F", border = NA, xlab = "Campaign stops, 2024",
        main = "Where the candidates spent their days")
```

Where the candidates spent their days



```
tot <- sum(visits$visits)
top7 <- sum(head(sort(visits$visits, decreasing = TRUE), 7))
round(100 * top7 / tot, 1)
```

```
#> [1] 83.8
```

Seven states absorbed 83.8% of the entire 2024 campaign. Pennsylvania alone took 80 stops — one every three days for eight months.

Part 3: Where they did not go

Join the visits onto the results file, keeping every state:

```
m <- merge(st, visits, by = "state", all.x = TRUE)
m$visits[is.na(m$visits)] <- 0
nrow(m)
```

```
#> [1] 51
```

```
sum(m$visits == 0)
```

```
#> [1] 25
```

Twenty-five states and territories — half the country — received not one visit. Not one, in eight months, from any of the five people on the two tickets.

Look at who was ignored:

```
m$absmargin <- abs(m$margin)
big_ignored <- m[m$ev >= 10 & m$visits <= 2, c("state", "ev", "absmargin", "visits")]
big_ignored[order(-big_ignored$ev), ]
```

```
#>      state ev absmargin visits
#> 5   California 54    20.14      2
#> 36    Ohio 17    11.35      2
#> 31  New Jersey 14     5.91      2
#> 48  Washington 12    18.20      0
#> 15    Indiana 11    18.96      1
#> 22 Massachusetts 11    25.20      2
#> 43    Tennessee 11    29.72      1
#> 6     Colorado 10    10.99      1
#> 21    Maryland 10    28.54      1
#> 26    Missouri 10    18.41      0
```

```
m[m$state %in% c("California", "Texas", "Nevada"),
  c("state", "ev", "absmargin", "visits")]
```

```
#>      state ev absmargin visits
#> 5   California 54    20.14      2
#> 29    Nevada  6     3.10     25
#> 44    Texas 40    13.68      5
```

California has 54 electoral votes and got 2 visits. Nevada has 6 and got 25. Nevada has roughly one-ninth of California's electoral weight and received more than twelve times the attention.

Per electoral vote, Nevada got about **a hundred and thirteen times** what California did.

Those are two states. Now put a number on the whole distribution.

Before you run the next chunk, write a percentage down. Line all 51 jurisdictions up from least-visited to most-visited *per electoral vote*, then walk down that list — the ones that got nothing first, then the barely-visited ones — until the electoral votes you have collected add up to half the country. **What share of the 377 campaign stops did that half receive?** Commit to a number now, while the tables above are all you have to go on.

```
o <- m[order(m$visits / m$ev, m$visits), ] # least attention per electoral vote first
ec <- cumsum(o$ev) / sum(o$ev) # running share of the electoral college
vis <- cumsum(o$visits) / sum(o$visits) # running share of all campaign stops

half <- approx(c(0, ec), c(0, vis), 0.5)$y
round(100 * half, 1)
```

```
#> [1] 2.5
```

2.5% — about 9 stops out of 377. Most people write down something in the teens, and the reasoning is sound as far as it goes: the 25 jurisdictions with no stops at all are only 27.1% of the electoral college, so the states sitting just above them ought to have picked up a respectable remainder. They did not. Half the country, weighted the way the Constitution weights it, was worth 9 stops in eight months.

Part 4: Is it about size?

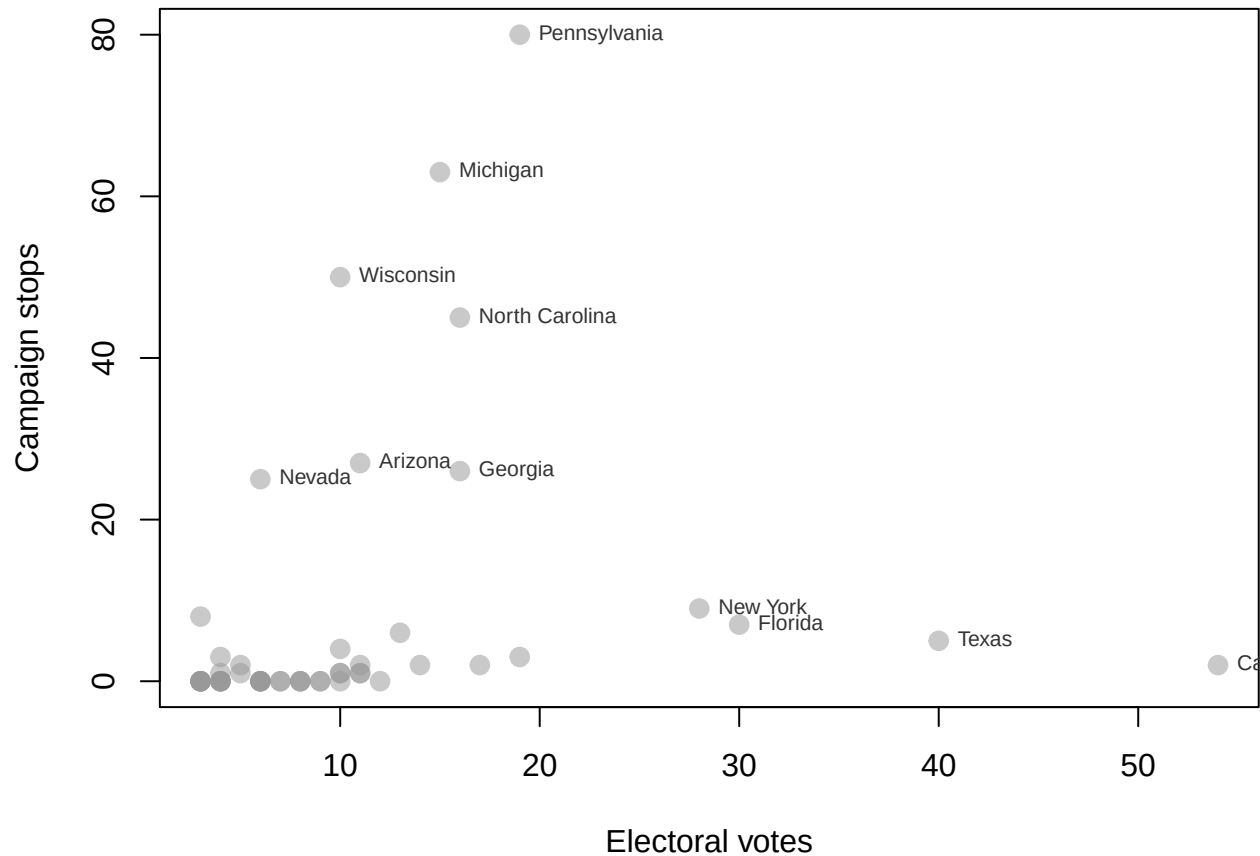
The obvious hypothesis is that candidates go where the votes are. Test it:

```
round(cor(m$visits, m$ev), 3)
```

```
#> [1] 0.207
```

```
plot(m$ev, m$visits, pch = 19, col = "#99999988", cex = 1.3,
     xlab = "Electoral votes", ylab = "Campaign stops",
     main = "Size barely predicts attention")
big <- m[m$ev >= 28 | m$visits >= 20, ]
text(big$ev, big$visits, big$state, pos = 4, cex = 0.7, col = "grey20")
```

Size barely predicts attention



A correlation of **0.21**. Essentially nothing. The largest states in the country sit along the bottom of that plot with almost no visits, and a cluster of medium-sized states sits far above them.

Size is not the answer.

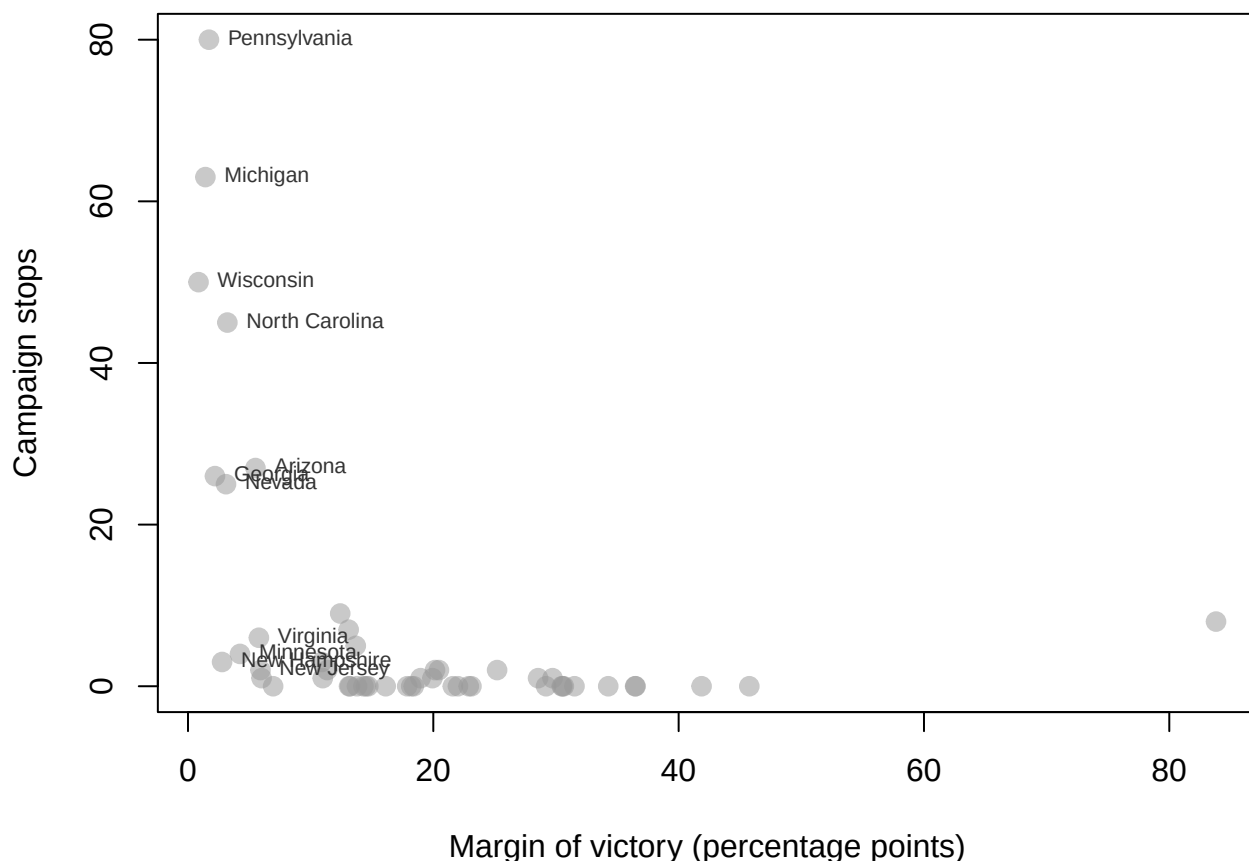
Part 5: Is it about closeness?

```
round(cor(m$visits, m$absmargin), 3)
```

```
#> [1] -0.408
```

```
plot(m$absmargin, m$visits, pch = 19, col = "#99999988", cex = 1.3,  
     xlab = "Margin of victory (percentage points)", ylab = "Campaign stops",  
     main = "Closeness predicts attention better - but not perfectly")  
lab <- m[m$visits >= 20 | (m$absmargin < 6 & m$visits < 10), ]  
text(lab$absmargin, lab$visits, lab$state, pos = 4, cex = 0.7, col = "grey20")
```

Closeness predicts attention better – but not perfectly



—0.41. Better, and in the right direction: the closer the state, the more visits. But it is not overwhelming either, and the plot shows why — New Hampshire and Minnesota were both close and were both largely left alone.

So neither explains it. Now put them together.

```
close <- m[m$absmargin < 5, ]
round(cor(close$visits, close$ev), 3)           # among close states, does size matter?
```

```
#> [1] 0.735
```

```
large <- m[m$ev >= 15, ]
round(cor(large$visits, large$absmargin), 3)   # among large states, does closeness matter?
```

```
#> [1] -0.834
```

There it is. Among the eight states decided by under five points, the correlation between size and visits is **+0.74** — strong. Among the ten largest states, the correlation between margin and visits is **−0.83** — very strong.

Neither size nor closeness works on its own. **Both work, conditionally.** A campaign is not asking “where are the voters?” or “where is it close?” It is asking a single question that multiplies the two: *where might my time change who wins?* A big safe state fails that test. A tiny close state passes it barely. A big close state is worth eighty visits.

This is the whole logic of the Electoral College expressed as a travel schedule.

Part 6: Three things to notice before you trust any of this

The tracker stops on 4 November — the day before the election.

```
max(v$date)
```

```
#> [1] "2024-11-04"
```

```
sum(v$date >= as.Date("2024-10-25"))
```

```
#> [1] 100
```

That is one day short, not four: the final weekend is here, and 100 of the 377 stops fall in the last eleven days. Every number is still a **floor**, but a close one.

It was not always. **The source kept moving after we first read it.**

```
ap <- read.csv("data/ap_snapshots.csv", stringsAsFactors = FALSE)
ap
```

```
#>   version rows last_event states rows_in_shared_window
#> 1      50  324 2024-11-01     25                      324
#> 2      54  377 2024-11-04     26                      339
```

The AP tracker lives at a URL carrying a **version number**, and AP kept publishing new versions after the election. This lab was first built against version 50, which ended on 1 November with 324 stops. It is now built against version 54. Two things changed, and only one is obvious.

The tracker **gained the ending** — 38 stops on 2–4 November that version 50 could not show. But it also **rewrote a period it had already covered**: for the window both versions claim, up to 1 November, version 50 has 324 stops and version 54 has 339. Fifteen events were added to a stretch the file already claimed to describe, and one of them put a new state on the map — New Mexico went from zero visits to 1, changing the count in Part 3 by a whole state.

A public dataset with no statutory deadline has no moment at which it is finished. The census closes; an FEC filing has a due date. This has neither: it is complete when a newsroom stops updating it, and nothing announces that day. **A citation to this source without a version number does not identify what was read.**

The candidate changed mid-dataset.

```
tapply(v$date, v$candidate, range)
```

```
#> $Biden
#> [1] "2024-03-13" "2024-07-16"
#>
#> $Harris
```



```
#> [1] "2024-07-23" "2024-11-04"
#>
#> $Trump
#> [1] "2024-03-16" "2024-11-04"
#>
#> $Vance
#> [1] "2024-07-22" "2024-11-04"
#>
#> $Walz
#> [1] "2024-08-13" "2024-11-04"
```

Biden’s last logged stop and Harris’s first are three weeks apart in July. If you were comparing “the Democratic candidate” to “the Republican candidate” over time, you would be comparing a continuous campaign against a broken one. As whole tickets, though, the two sides are almost perfectly matched:

```
c(Democratic = sum(v$candidate %in% c("Biden", "Harris", "Walz")),
  Republican = sum(v$candidate %in% c("Trump", "Vance")))
```

```
#> Democratic Republican
#>          188          189
```

One state defies the entire pattern.

```
m[m$state == "District of Columbia", c("state", "ev", "absmargin", "visits")]
```

```
#>           state ev absmargin visits
#> 9 District of Columbia 3      83.81      8
```

The District of Columbia was decided by **84 points** and still logged 8 visits. Nobody was campaigning there for votes. It is where the candidates already were — where the fundraisers, the television studios, and in one case the office are. A row can satisfy your filter and mean something entirely different from the rest.

Part 7: On your own

This is the part you turn in. **You are not being asked to write code.** Each option is a working chunk with one thing to change, or nothing at all.

Pick **one**.

Option A — Inside one state

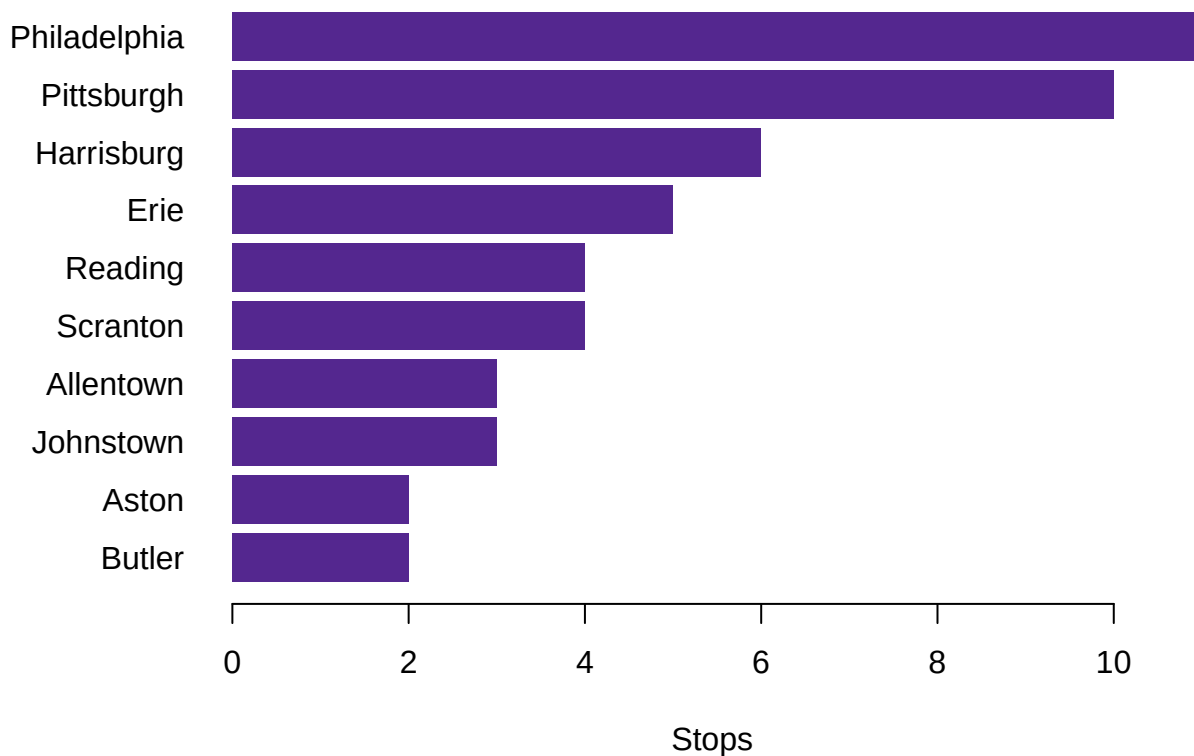
```
this_state <- "Pennsylvania"      # <- change to any state in the data

s <- v[v$state == this_state, ]
cities <- sort(table(s$city), decreasing = TRUE)
head(cities, 10)
```

```
#>
#> Philadelphia    Pittsburgh    Harrisburg      Erie      Reading    Scranton
#>           11           10           6           5           4           4
#>    Allentown    Johnstown      Aston      Butler
#>           3           3           2           2
```

```
top_c <- head(cities, 10)
par(mar = c(5, 10, 4, 2))
barplot(rev(as.numeric(top_c)), names.arg = rev(names(top_c)), horiz = TRUE,
        las = 1, col = "#54278F", border = NA, xlab = "Stops",
        main = paste("Campaign stops in", this_state))
```

Campaign stops in Pennsylvania



```
nrow(s)
```

```
#> [1] 80
```

```
length(unique(s$city))
```

```
#> [1] 36
```

Write about: Pennsylvania's 80 stops are spread across 36 towns, not concentrated in Philadelphia. Why would a campaign go to Erie or Johnstown rather than hold more events in the largest city? Think about who is already going to vote and who might be persuaded — and about the fact that a campaign stop generates *local* news coverage. Then try a smaller battleground like Nevada and say whether the same logic is visible.

Option B — Attention per voter

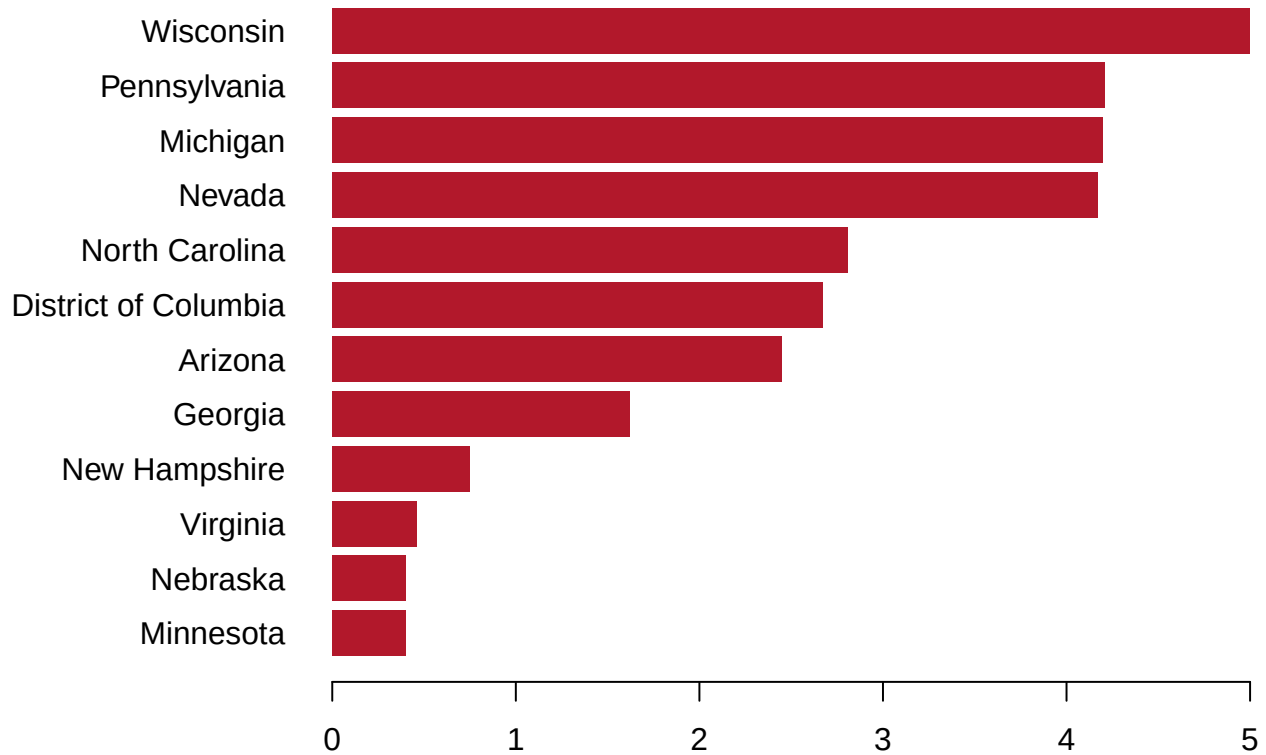
```
m$per_ev <- round(m$visits / m$ev, 2)

top_pe <- head(m[order(-m$per_ev), c("state", "ev", "absmargin", "visits", "per_ev")], 12)
top_pe
```

```
#>           state ev absmargin visits per_ev
#> 50      Wisconsin 10      0.86     50   5.00
#> 39    Pennsylvania 19      1.71     80   4.21
#> 23         Michigan 15      1.42     63   4.20
#> 29         Nevada  6      3.10     25   4.17
#> 34    North Carolina 16      3.21     45   2.81
#> 9  District of Columbia 3    83.81      8   2.67
#> 3           Arizona 11      5.50     27   2.45
#> 11          Georgia 16      2.20     26   1.62
#> 30    New Hampshire  4      2.78      3   0.75
#> 47         Virginia 13      5.78      6   0.46
#> 24         Minnesota 10      4.24      4   0.40
#> 28         Nebraska  5     20.46      2   0.40
```

```
seen <- m[m$visits > 0, ]
par(mar = c(5, 11, 4, 2))
sel <- head(seen[order(-seen$per_ev), ], 12)
sel <- sel[order(sel$per_ev), ]
barplot(sel$per_ev, names.arg = sel$state, horiz = TRUE, las = 1,
        col = "#B2182B", border = NA, xlab = "Campaign stops per electoral vote",
        main = "Attention per unit of electoral weight")
```

Attention per unit of electoral weight



Campaign stops per electoral vote

Write about: which states are most over-attended relative to their actual electoral weight, and which of the fifty are most under-attended? Wisconsin and Nevada should be near the top. Then the question worth arguing about: the Electoral College is often defended on the grounds that it forces candidates to pay attention to small states. Does this table support that defence, undercut it, or split the difference? Be specific about *which* small states.

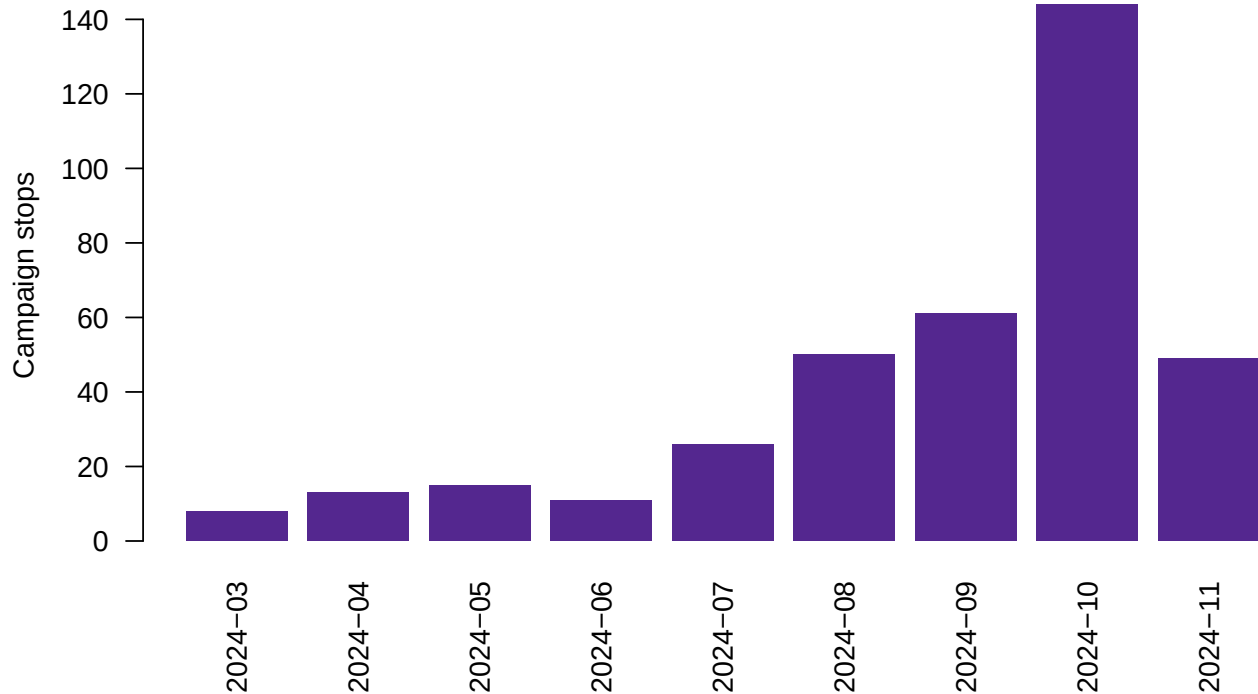
Option C — The shape of the calendar

```
v$month <- format(v$date, "%Y-%m")
by_month <- table(v$month)
by_month
```

```
#>
#> 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10 2024-11
#>      8      13      15      11      26      50      61     144      49
```

```
barplot(as.numeric(by_month), names.arg = names(by_month), las = 2,
        col = "#54278F", border = NA, ylab = "Campaign stops",
        main = "When the campaign happened")
```

When the campaign happened



```
# How concentrated is the whole thing in the final stretch?  
round(100 * sum(v$date >= as.Date("2024-09-01")) / nrow(v), 1)
```

```
#> [1] 67.4
```

Write about: describe the shape. What share of an eight-month campaign happened in the last two months? Then account for two distortions before you draw any conclusion: the tracker stops on 4 November, the day before the election, and the Democratic candidate changed in late July. The first now shaves off one day rather than a whole final weekend — does it still matter, and how much? The second is larger than it looks. A good answer separates what the campaign did from what the dataset can see, and says which distortion is actually doing work.

Option D — Ask your own question

Compare the two tickets' maps. Look at what `event_type` and `notes` contain. Find the states one side visited and the other did not. Follow whatever nags.

What to submit. Knit to HTML or PDF and upload to Canvas, with your chosen chunk and **two or three paragraphs** of interpretation.

The writing is the assignment. I am not grading your code — it is my code. I am grading whether you can look at a travel schedule and explain what it reveals about how American elections are actually decided.

Where the data came from

Associated Press, “Tracking the 2024 presidential campaign trail,” **version 54** (<https://interactives.ap.org/embeds/ziEfd/54/>). AP logged public campaign appearances by both tickets from March 2024 to 4 November 2024.

Four limits, all of which shape what you can say:

- **It ends one day early.** Election day itself is absent.
- **It is versioned, and the version moved.** This lab was built against version 50 (324 stops, ending 1 November) before version 54 replaced it; 15 of the extra stops were backfilled into a window version 50 already claimed to cover. See `data/ap_snapshots.csv`, and note that nothing guarantees 54 is the last.
- **It counts events, not effort.** A stadium rally and a stop at a diner are both one row. Nothing here measures crowd size, time spent, or money spent.
- **It is a news organisation’s log, not a record of a legal filing.** Unlike last week’s FEC data, nobody was required to report any of this. AP built it by watching. Coverage of the running mates is likely less complete than of the principals.

State results are the same file used in Weeks 1 and 2.

The script that downloaded, checked and trimmed all of this is `data/build-data.R`. It verifies the state-name join before the lab depends on it — a habit you should now expect to see.